



Sustainable Energy Ecosystem (SEE) – Platform Scoping & Costing Proposal

Full-Stack Design, Development & Delivery of the SEE Platform

POWERED BY THE REISIGER APEX SUITE

CONFIDENTIALITY NOTICE

This document and all information contained herein are confidential and intended solely for the recipient named herein. The contents of this document, in whole or in part, are proprietary to Reisiger Engineering and MW-GS Pty Ltd / Phoenix Energy Solutions and may not be disclosed, copied, distributed, or used for any purpose other than the evaluation of this document without the express prior written consent of Reisiger Engineering.



EXECUTIVE SUMMARY

The Opportunity

The Sustainable Energy Ecosystem (SEE) is an ambitious platform designed to become the operating system for renewable energy project development in Southern Africa. It connects four actors — **Contractors (EPCs)**, **Service Providers**, **End-Clients**, and **Platform Administrators** — into a single ecosystem that governs project lifecycle, compliance, commerce, monitoring, and intelligence.

The platform is a 50/50 joint venture between MW-GS Pty Ltd and Phoenix Energy Solutions. The Functional Requirements Document (v1, February 2026) and the S.E.E. Journey architecture map describe a comprehensive vision spanning project management, milestone governance, dual marketplaces (services + hardware), a token-based rewards economy, multi-brand O&M monitoring, AI-powered verification, prescriptive maintenance, and an intelligent assistant.

The Challenge

The full SEE vision is a **R4.6M+ platform build**. Attempting to deliver the entire scope in a single phase would create excessive delivery risk, delay time-to-market, and consume capital before the core value proposition is validated with real users.

Our Approach — Phased Delivery

This proposal breaks the SEE platform into **six delivery phases**, each producing a functional, deployable increment. Phase 1 delivers a working MVP that validates the core value proposition — contractors loading and managing projects with milestone governance. Each subsequent phase layers on additional capability: verification, marketplaces, token economy, O&M monitoring, and AI intelligence.

MW-GS and Phoenix Energy Solutions can choose to proceed phase-by-phase, validating market traction before committing to the next investment. No phase requires commitment to later phases.

The APEX Advantage

Reisiger builds on the **APEX Suite** — a unified enterprise intelligence platform with seven modules. SEE benefits from APEX as a foundation layer, providing production-grade capabilities that would otherwise need to be built from scratch: notification delivery (Echo), AI/ML analytics (Quantum), data ingestion and normalisation (Zenith), report generation (Prism), and client admin portals (Forge). This reduces development time, lowers risk, and provides enterprise-grade capability from day one.

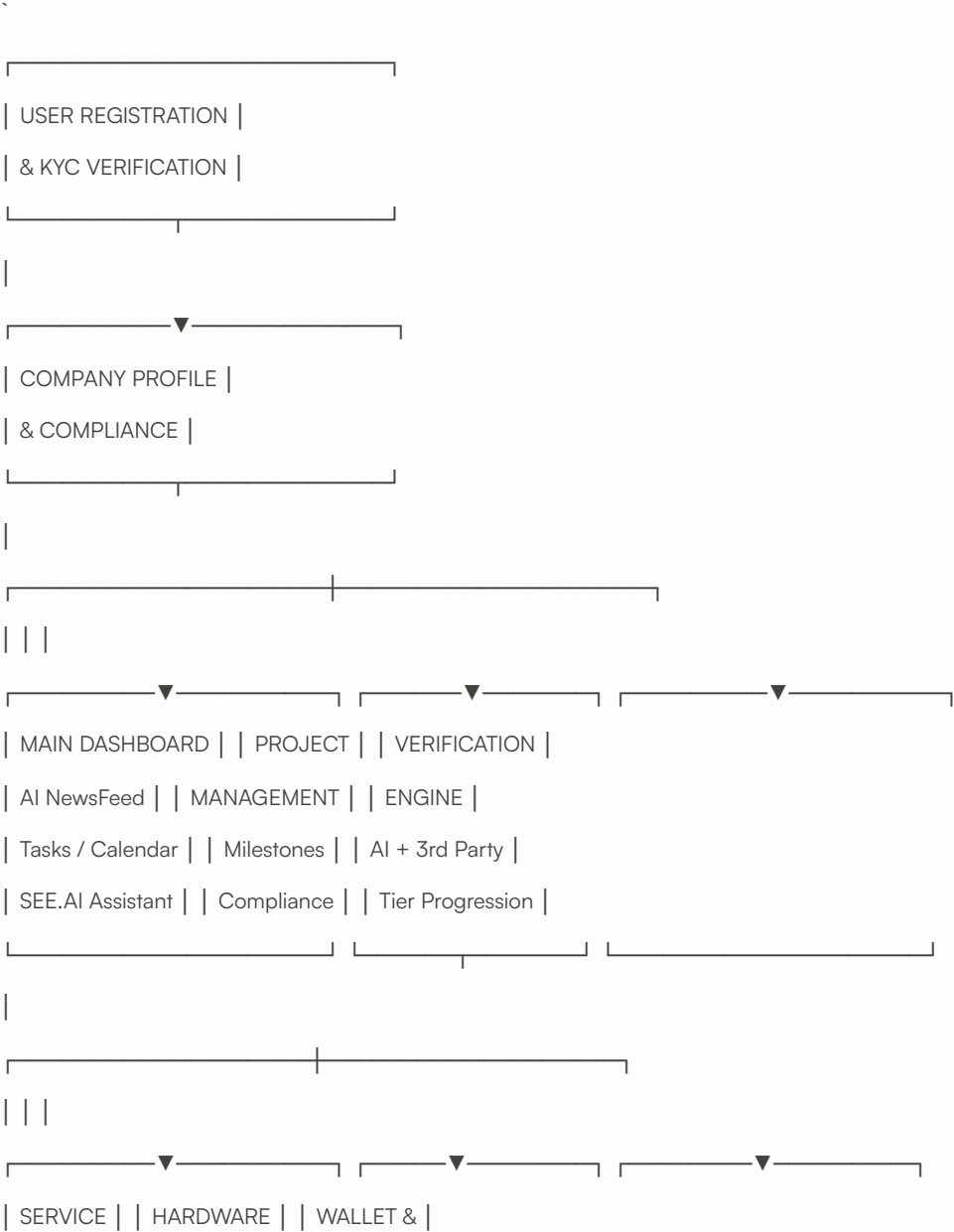
PLATFORM ARCHITECTURE – THE S.E.E. JOURNEY

The following architecture is derived from the S.E.E. Journey specification and represents the complete platform vision across all six phases.

Platform Actors

ACTOR	ROLE	KEY CAPABILITIES
Contractor (EPC)	Renewable energy contractors who load and manage projects	Project management, milestone tracking, O&M monitoring, sell projects
Service Provider	Professional service firms listed on the marketplace	Structural & civils, engineering, legal, logistics, finance & insurance
End-Client	Property/asset owners commissioning renewable energy installations	Project visibility, O&M access (outright purchase projects)
Platform Admin	SEE platform operators	User verification, compliance oversight, system configuration

Core Platform Modules



ACTION	TOKEN VALUE
Watch Tutorial	Earn 100 Tokens
Create a Project	Earn 100 Tokens
Service Request	Earn 100 Tokens
Complete Existing Project Upload	Earn 1,000 Tokens
AI Verification	Costs 1,000 Tokens
3rd Party Expert Verification	Costs 10,000 Tokens
Company Profile Generator	Costs 1,000 Tokens
Marketplace Cashback	Earn % Tokens based on Tier (e.g., 10% on Gold)

PHASE 1 – PLATFORM FOUNDATION & CORE PROJECT MANAGEMENT (MVP)

Goal: Deliver a working platform where contractors can register, create company profiles, load projects, track milestones, and achieve compliance — the core value loop that validates the SEE concept.

What Gets Built

1.1 Infrastructure & Architecture

ITEM	DESCRIPTION	HOURS
Cloud infrastructure setup	AWS infrastructure — VPC, subnets, security groups, RDS, S3, CloudFront	16
Database design & schema	Full relational schema for users, companies, projects, milestones, documents, tokens	20
CI/CD pipeline	GitHub Actions — build, test, deploy to staging and production	8
Environment configuration	Dev, staging, production environments with secrets management	8
API architecture & gateway	RESTful API design, API gateway, authentication middleware, rate limiting	12
Subtotal		64

1.2 Authentication & User Management

ITEM	DESCRIPTION	HOURS
Registration flow	Email registration + Google/Microsoft SSO	16
Email verification & password management	Verification emails, password reset, account security	8
KYC workflow engine	Multi-step verification: CIPC registration, VAT registration, Directors ID upload	28
KYC document upload & secure storage	Encrypted document storage, file type validation, virus scanning	8
Admin verification queue	Pending/Approved/Rejected workflow with admin review interface	16
Role-based access control	Contractor, Service Provider, Client, Admin roles with permission matrix	12
Subtotal		88

1.3 Company Profile Module

ITEM	DESCRIPTION	HOURS
Company profile creation & editing	Company details, contact information, operating regions	16
KYC management dashboard	Company Registrations, BEEE certificates, Licences & Certifications tracking	24
Bank details & statement upload	Secure bank detail capture and statement document storage	8
Profile settings & Corporate Identity Pack	Logo upload, brand colours, document templates	12
Compliance status tracking	Real-time compliance score based on KYC completeness	8
Subtotal		68

1.4 Project Management Core

ITEM	DESCRIPTION	HOURS
New project creation wizard	Step-by-step project creation with minimum required details	24
Existing project quick upload & import	Bulk import flow with incentive tracking (1,000 tokens for complete upload)	24
Project details management	Full project information — location, capacity, system type, client details	16
Milestone/Compliance template system	Pre-built milestone templates for Development, Construction, and O&M phases	28
Milestone generation engine	Auto-generate project milestones from selected templates	20
Document upload, versioning & compliance	Per-milestone document upload with version control and compliance status	20
Milestone progress tracking	Visual progress indicators, completion status, compliance percentage	12
Subtotal		144

1.5 Main Dashboard

ITEM	DESCRIPTION	HOURS
Dashboard layout & navigation	Responsive dashboard with sidebar navigation and widget layout	16
Tasks / To-Do List	Task management with project-linked tasks, due dates, assignments	16
Calendar events	Calendar view with milestone deadlines, verification dates, project events	16
Basic notification feed	In-app notification centre for project updates, KYC status, system messages	12
Plant notifications / suggestions (basic)	Basic notification placeholders for future O&M integration	8
Subtotal		68

1.6 Admin Panel (Foundation)

ITEM	DESCRIPTION	HOURS
User management	User listing, search, status management, role assignment	16
KYC verification queue & approval workflow	Admin review interface for KYC submissions with approve/reject/request-info actions	16
Project oversight	Admin view of all projects, compliance status, milestone progress	12
System configuration	Platform settings, template management, notification configuration	12
Subtotal		56

1.7 Onboarding & Tutorial System

ITEM	DESCRIPTION	HOURS
Tutorial / Demo system	Guided tutorial flow (Watch Tutorial, Create a Project, Service Request)	12
Onboarding wizard	First-time user experience with progressive disclosure	8
Token incentive engine (foundation)	Award 100 tokens per tutorial action, 1,000 for existing project upload	12
Subtotal		32

1.8 UX/UI Design

ITEM	DESCRIPTION	HOURS
UI/UX design & prototyping	Full design system, component library, user flow prototypes	24
Responsive design implementation	Mobile-first responsive implementation across all Phase 1 screens	16
Design review & iteration	Stakeholder review cycles with Phoenix Energy Solutions and MW-GS	8
Subtotal		48

Phase 1 – APEX Acceleration

APEX MODULE	ROLE IN PHASE 1
Forge	Admin dashboard, user management, KYC verification queue — reduces custom admin panel build by ~40%
Echo	Notification delivery engine — email, in-app, and push notifications without building custom notification infrastructure

Phase 1 – Commercial Summary

ITEM	DETAIL
Hours	568
Rate	R2,250 per hour, excluding VAT
Cost	R1,278,000 excl. VAT
Duration	~7 weeks (2 dedicated engineers, 80 productive hours/week)
Deposit	R383,400 (30%) payable on scope approval — triggers commencement
Balance	R894,600 payable monthly based on tracked hours until completion
Deliverable	Working platform: registration, KYC, company profiles, project management with milestones, admin panel, basic dashboard
Dependencies	Hosting account credentials, Phoenix Energy Solutions design review participation



PHASE 2 — VERIFICATION ENGINE & TIER SYSTEM

Goal: Enable the compliance and quality layer that differentiates SEE from every other project management tool — AI-powered verification, 3rd party expert review, Gold Standard certification, and tier progression that rewards compliant contractors.

What Gets Built

2.1 AI Verification Agent

ITEM	DESCRIPTION	HOURS
Document analysis engine	AI-powered analysis of uploaded milestone documents for compliance verification	36
Compliance rule engine	Configurable rule sets per milestone type — what constitutes a compliant submission	24
Verification result management	Pass/fail/pending results with detailed feedback and remediation guidance	16
Auto-Gold verification for marketplace artifacts	Milestones acquired through marketplace Service Providers auto-verified as Gold	12
Subtotal		88

2.2 3rd Party Expert Verification

ITEM	DESCRIPTION	HOURS
Expert onboarding & management	Expert registration, credentialing, specialisation tagging, availability management	20
Verification assignment workflow	Match verification requests to qualified experts, assignment notifications	20
Verification form system	Colour-coded quality rating (colour refers to quality of independent verification)	16
Expert review dashboard	Expert-facing interface for reviewing submissions and submitting verification reports	16
Subtotal		72

2.3 Tier Progression System

ITEM	DESCRIPTION	HOURS
Tier rules engine	Configurable progression rules: Bronze → Silver → Gold → Platinum based on compliant project count	24
Tier progression tracking	Real-time tier status, progress towards next tier, compliant project counting	16
Tier-based feature gating	Restrict access to leads, marketplace features, and cashback rates by tier	16
Progress visualisation	Tier badge display, progress bars, achievement milestones	8
Subtotal		64

2.4 Gold Standard Certification

ITEM	DESCRIPTION	HOURS
Compliance criteria engine	Define and manage Gold Standard requirements per project type	20
Certificate generation	Branded Gold Standard certificates for fully compliant projects	12
Project strength & weight scoring	Higher-tier projects receive greater visibility and trust weighting in the platform	12
Subtotal		44

2.5 Non-Compliant Project Handling

ITEM	DESCRIPTION	HOURS
Compliance notification engine	Persistent reminders for unverified/non-compliant projects until verified	12
Compliance upgrade prompts	Contextual prompts showing benefits of compliance and tier progression	8
Subtotal		20

2.6 Testing & QA

ITEM	DESCRIPTION	HOURS
Verification accuracy testing	End-to-end testing of AI and expert verification workflows	12
Tier progression integration testing	Scenario testing for tier advancement, feature gating, edge cases	8
Subtotal		20

Phase 2 – APEX Acceleration

APEX MODULE	ROLE IN PHASE 2
Quantum	Powers the AI Verification Agent — document analysis, compliance rule evaluation, and confidence scoring
Vector	Verification integrity — ensures AI verification claims are grounded against source compliance standards
Prism	Generates Gold Standard certificates and compliance reports as branded PDF outputs

Phase 2 – Commercial Summary

ITEM	DETAIL
Hours	308
Rate	R2,250 per hour, excluding VAT
Cost	R693,000 excl. VAT
Duration	~4 weeks (2 dedicated engineers)
Deposit	R207,900 (30%)
Balance	R485,100
Deliverable	AI verification, 3rd party expert verification, tier progression (Bronze-Platinum), Gold Standard certification, compliance enforcement
Dependencies	Phase 1 complete; compliance rule definitions from Phoenix Energy Solutions



PHASE 3 — SERVICE & HARDWARE MARKETPLACES

Goal: Open the commerce layer — a dual marketplace where contractors source professional services and hardware, service providers list their capabilities, and the platform earns commission on every transaction.

What Gets Built

3.1 Service Marketplace

ITEM	DESCRIPTION	HOURS
Category management	5 categories: Structural & Civils, Pr. Engineering Services, Legal, Logistics & Plant Hire, Finance + Insurance	12
Service provider onboarding & listing	Registration, verification, capability profiles, portfolio showcase	28
Service listing creation & management	Service descriptions, pricing models, availability, geographic coverage	20
Search, filtering & discovery	Category browsing, keyword search, location filtering, rating sort	24
Service request & quoting workflow	Request submission, quote receipt, comparison, acceptance, milestone linking	32
Review & rating system	Post-service reviews, star ratings, verified transaction badges	16
Service provider dashboard	Earnings, active requests, quote pipeline, performance metrics	20
Listing fee management	Service provider subscription/listing fees, billing, renewal reminders	12
Auto-Gold milestone verification	Milestones/artifacts acquired through marketplace transactions auto-verified as Gold	12
Subtotal		176

3.2 Hardware Marketplace

ITEM	DESCRIPTION	HOURS
Product catalogue management	Product entries with specifications, images, pricing, availability	20
Category structure	5 categories: Solar Panels, Batteries, Inverters, Generators, Accessories	8
Product listing, search & filtering	Browse, search, filter by brand/capacity/price, comparison view	24
Order management & fulfilment	Order placement, status tracking, delivery coordination, confirmation	28
Supplier management (direct partners)	Partner supplier onboarding, inventory sync, direct fulfilment	20
Affiliate link management	Non-partnered supplier products with affiliate links or manual order flow	12
Hardware marketplace dashboard	Sales metrics, order pipeline, supplier performance, inventory levels	12
Subtotal		124

3.3 Platform Fee Engine

ITEM	DESCRIPTION	HOURS
Payment gateway fee calculation	Transparent calculation and display of payment gateway fees	12
Platform fee engine	Configurable X% platform fee based on user's payment plan/tier	16
Fee reporting & analytics	Revenue tracking dashboard for platform operators	8
Subtotal		36

3.4 Testing & QA

ITEM	DESCRIPTION	HOURS
Marketplace integration testing	End-to-end service request and hardware order flows	12
Transaction flow testing	Payment processing, fee calculation, edge case handling	8
Subtotal		20

Phase 3 – APEX Acceleration

APEX MODULE	ROLE IN PHASE 3
Forge	Service provider dashboard and admin marketplace management interface
Echo	Transaction notifications — quote received, order placed, delivery updates, review reminders
Prism	Invoice and receipt generation for marketplace transactions

Phase 3 – Commercial Summary

ITEM	DETAIL
Hours	356
Rate	R2,250 per hour, excluding VAT
Cost	R801,000 excl. VAT
Duration	~5 weeks (2 dedicated engineers)
Deposit	R240,300 (30%)
Balance	R560,700
Deliverable	Live service marketplace (5 categories), hardware marketplace (5 categories), platform fee engine, service provider onboarding, review system
Dependencies	Phase 1 complete; service provider partners identified for launch; hardware supplier partnerships or affiliate agreements established



PHASE 4 — TOKEN ECONOMY, WALLET & PAYMENTS

Goal: Deploy the full financial infrastructure — token wallet, payment gateway, subscription billing, cashback rewards, and the "Discovery Miles" token-as-discount model that drives platform engagement and retention.

What Gets Built

4.1 Wallet System

ITEM	DESCRIPTION	HOURS
Token ledger & balance management	Double-entry token ledger with real-time balance, earn/spend tracking, audit trail	20
Token purchase flow	Fiat-to-token conversion with configurable exchange rate, multiple payment methods	24
Token-as-discount engine	"Discovery Miles" model — tokens allocated at checkout to reduce cash cost on services and hardware	28
Transaction history & audit	Full transaction history with filtering, export, and compliance audit trail	12
Subtotal		84

4.2 Payment Gateway Integration

ITEM	DESCRIPTION	HOURS
Payment processing integration	PayFast (primary, SA market) integration for card payments, EFT, instant EFT	28
Payment reconciliation	Automated reconciliation between gateway transactions and platform ledger	16
Refund & dispute handling	Refund workflow, dispute escalation, chargeback management	16
PCI compliance measures	Tokenised card storage, secure payment pages, audit logging	8
Subtotal		68

4.3 Rewards Engine

ITEM	DESCRIPTION	HOURS
Cashback calculation	Tier-based cashback percentages (e.g., 10% token cashback on Gold Tier transactions)	16
Token distribution rules engine	Configurable earn rules — tutorial completion, project upload, marketplace transactions	16
Progress & Rewards Tracker	User-facing rewards dashboard showing earn history, cashback earned, tier benefits	12
Incentive trigger system	Automated token distribution on qualifying actions (tutorials, uploads, transactions)	12
Subtotal		56

4.4 Subscription Billing

ITEM	DESCRIPTION	HOURS
Subscription tier management	Freemium / Pro / Enterprise tier definitions with feature matrices	20
Recurring billing engine	Monthly subscription billing, payment retry logic, grace periods, cancellation	20
Feature gating per subscription	Runtime feature flags tied to subscription tier — Pro gets 20 free AI verifications, Enterprise gets unlimited	12
Upgrade/downgrade workflows	Self-service tier changes with prorated billing	8
Subtotal		60

4.5 Invoicing & Financial Reporting

ITEM	DESCRIPTION	HOURS
Invoice generation	Automated invoice generation for subscriptions, token purchases, marketplace transactions	16
Purchase history	User-facing purchase history with invoice download	8
Financial reporting dashboard	Platform revenue reporting — subscriptions, marketplace fees, token sales, O&M licences	12
Subtotal		36

4.6 Testing & QA

ITEM	DESCRIPTION	HOURS
Payment flow testing	End-to-end payment, refund, subscription, and token purchase flows	12
Token economy balancing	Stress testing earn/spend ratios, edge cases, negative balance prevention	8
Subtotal		20

Phase 4 – APEX Acceleration

APEX MODULE	ROLE IN PHASE 4
Forge	Financial reporting dashboard, subscription management admin interface
Prism	Invoice and receipt PDF generation
Echo	Payment confirmations, subscription reminders, cashback notifications

Phase 4 – Commercial Summary

ITEM	DETAIL
Hours	324
Rate	R2,250 per hour, excluding VAT
Cost	R729,000 excl. VAT
Duration	~4 weeks (2 dedicated engineers)
Deposit	R218,700 (30%)
Balance	R510,300
Deliverable	Token wallet, payment gateway (PayFast), subscription billing (Freemium/Pro/Enterprise), cashback rewards engine, invoicing, financial reporting
Dependencies	Phase 1 complete; PayFast merchant account established; token pricing and cashback rates finalised by MW-GS/Phoenix Energy



PHASE 5 — O&M MONITORING & PRESCRIPTIVE INTELLIGENCE

Goal: Deploy the ongoing-value layer — multi-brand inverter data integration, real-time plant monitoring dashboards, tiered O&M licensing, and AI-powered prescriptive maintenance that transforms SEE from a project management tool into a long-term revenue platform.

What Gets Built

5.1 Multi-Brand Inverter Data Integration

ITEM	DESCRIPTION	HOURS
WEG inverter API integration	WEG monitoring API connectivity, data mapping, authentication	28
Victron (VRM) API integration	Victron Remote Management API connectivity and data mapping	24
SunSynk API integration	SunSynk monitoring portal API connectivity and data mapping	24
Deye API integration	Deye cloud monitoring API connectivity and data mapping	20
Data normalisation layer	Unified data model across all brands — power output, battery state, grid status, consumption, errors	16
Real-time data pipeline	Streaming data ingestion with configurable polling intervals, data validation, and storage	16
Subtotal		128

5.2 Plant Monitoring Dashboard

ITEM	DESCRIPTION	HOURS
Plant overview & performance metrics	Real-time system status, power generation, consumption, battery state, grid status	24
Real-time data visualisation	Live charts, gauges, and status indicators with auto-refresh	20
Historical performance analytics	Time-series analysis — daily/weekly/monthly/annual generation, trends, comparisons	20
Alert & notification system	Configurable thresholds for performance degradation, faults, communication loss	16
Multi-plant portfolio view	Aggregated view across all plants for contractors managing multiple installations	12
Subtotal		92

5.3 O&M Licence Management

ITEM	DESCRIPTION	HOURS
Tiered licence system	Basic Licence (monitoring only), Premium Licence (monitoring + alerts + reports), AI Licence (all + prescriptive maintenance)	24
Licence pricing engine	Dynamic pricing based on project size, value, location, and system configuration	16
Monthly / annual billing	Recurring licence billing with pro-rata calculations, renewal management	12
Feature gating per licence	Runtime feature flags — Basic sees dashboards, Premium gets alerts, AI gets prescriptive features	8
Subtotal		60

5.4 AI Prescriptive Maintenance (AI Licence)

ITEM	DESCRIPTION	HOURS
Automatic cleaning schedule generation	AI-generated panel cleaning schedules based on soiling rates, weather data, and performance curves	24
Smart replacement recommendations	Component degradation analysis with replacement timing and ROI calculations	24
Configuration optimisation	Auto-switching between self-consumption and back-up modes based on grid status, tariff schedules, and anticipated load-shedding	24
Load-shedding anticipation engine	Integration with load-shedding schedules (Eskom SE Push API or equivalent) for proactive mode switching	16
Subtotal		88

5.5 Testing & QA

ITEM	DESCRIPTION	HOURS
Multi-brand integration testing	Validate data accuracy across all four inverter brands with test installations	16
Prescriptive accuracy validation	Validate AI recommendations against real-world maintenance outcomes	8
Subtotal		24

Phase 5 — APEX Acceleration

APEX MODULE	ROLE IN PHASE 5
Zenith	Multi-brand inverter data ingestion and normalisation — the core data pipeline for O&M monitoring
Quantum	Powers prescriptive maintenance — cleaning schedules, replacement recommendations, configuration optimisation
Echo	O&M alert delivery — performance degradation warnings, fault notifications, cleaning reminders
Prism	O&M performance reports — branded monthly/quarterly plant performance PDFs

Phase 5 – Commercial Summary

ITEM	DETAIL
Hours	392
Rate	R2,250 per hour, excluding VAT
Cost	R882,000 excl. VAT
Duration	~5 weeks (2 dedicated engineers)
Deposit	R264,600 (30%)
Balance	R617,400
Deliverable	Multi-brand inverter integration (WEG, Victron, SunSynk, Deye), plant monitoring dashboards, tiered O&M licensing (Basic/Premium/AI), AI prescriptive maintenance
Dependencies	Phase 1 complete; API access/credentials from at least one inverter brand for initial integration; load-shedding API access



PHASE 6 — SEE.AI & ADVANCED INTELLIGENCE

Goal: Deploy the intelligence layer across the entire platform — the SEE.AI Assistant with full portfolio knowledge, AI-powered newsfeed, Company Profile and Proposal generators, multi-language customer support, and the "Sell My Project" marketplace.

What Gets Built

6.1 SEE.AI Assistant

ITEM	DESCRIPTION	HOURS
LLM integration & prompt engineering	Foundation model integration, domain-specific system prompts, conversation management	36
Portfolio knowledge base	Index entire platform data — projects, milestones, compliance, O&M data — for AI-accessible retrieval	28
Contextual suggestions engine	Proactive suggestions as users navigate milestones, compliance requirements, and platform features	24
Multi-language support	Full conversational support across English, Afrikaans, Zulu, and other SA languages	20
Customer call centre capability	AI-powered first-line support — platform help, feature guidance, troubleshooting	20
Usage limits by package	SEE.AI usage gating tied to subscription tier (Freemium/Pro/Enterprise)	8
Subtotal		136

6.2 AI NewsFeed

ITEM	DESCRIPTION	HOURS
Industry news aggregation	Automated ingestion from renewable energy news sources, regulatory updates, industry publications	16
AI curation & personalisation	AI-powered content ranking based on user's project types, location, and activity patterns	20
Feed UI & engagement	News feed within main dashboard, read/save/share actions, engagement tracking	8
Subtotal		44

6.3 AI Content Generation

ITEM	DESCRIPTION	HOURS
Company Profile Generator	AI-generated company profiles from KYC data and project history (1,000 Tokens)	20
Proposal generator	AI-generated project proposals using company portfolio, project data, and templates	20
Company Portfolio generator	Aggregated portfolio document showcasing completed projects, certifications, and capabilities	12
Subtotal		52

6.4 Sell My Project

ITEM	DESCRIPTION	HOURS
Project valuation engine	AI-assisted project valuation based on system size, generation data, contract terms, location	28
Project listing & discovery	Marketplace for projects available for sale with search, filtering, and detailed project views	20
Buyer/seller workflow	PPA/Lease option: EPC can list; Outright purchase: client can list; negotiation, due diligence, and transfer workflow	20
Subtotal		68

6.5 WEG Product Intelligence

ITEM	DESCRIPTION	HOURS
WEG product recommendation engine	AI trained to recommend WEG products based on project specifications and requirements	12
Product comparison & specification matching	Compare WEG products against project requirements with AI-generated suitability scores	8
Subtotal		20

6.6 Testing & QA

ITEM	DESCRIPTION	HOURS
AI response quality testing	Evaluate accuracy, relevance, and safety of AI responses across use cases	12
Multi-language validation	Test conversational quality across all supported languages	8
Subtotal		20

Phase 6 — APEX Acceleration

APEX MODULE	ROLE IN PHASE 6
Quantum	Powers SEE.AI Assistant, NewsFeed curation, content generation, project valuation, WEG recommendations
Zenith	Portfolio knowledge base — indexes all platform data for AI-accessible retrieval
Prism	Company Profile PDF generation, Proposal output, Portfolio document generation
Vector	Ensures AI-generated content (profiles, proposals) is grounded against real platform data

Phase 6 – Commercial Summary

ITEM	DETAIL
Hours	340
Rate	R2,250 per hour, excluding VAT
Cost	R765,000 excl. VAT
Duration	~4 weeks (2 dedicated engineers)
Deposit	R229,500 (30%)
Balance	R535,500
Deliverable	SEE.AI Assistant with full portfolio knowledge, AI NewsFeed, Company Profile/Proposal/Portfolio generators, Sell My Project marketplace, WEG product intelligence, multi-language support
Dependencies	Phases 1-2 minimum; Phase 5 recommended for full O&M AI integration; WEG product catalogue and specifications



INVESTMENT SUMMARY

Full Platform Build — All Six Phases

PHASE	SCOPE	HOURS	DURATION	COST (EXCL. VAT)
Phase 1	Platform Foundation & Core MVP	568	~7 weeks	R1,278,000
Phase 2	Verification Engine & Tier System	308	~4 weeks	R693,000
Phase 3	Service & Hardware Marketplaces	356	~5 weeks	R801,000
Phase 4	Token Economy, Wallet & Payments	324	~4 weeks	R729,000
Phase 5	O&M Monitoring & Prescriptive AI	392	~5 weeks	R882,000
Phase 6	SEE, AI & Advanced Intelligence	340	~4 weeks	R765,000
Total	Full Platform	2,288	~29 weeks	R5,148,000

Rate: R2,250 per hour, excluding VAT

Resources: 2 dedicated engineers (80 productive hours per week)

Engagement model: Services pillar — Reisiger as technology partner

Billing Per Phase

PHASE	DEPOSIT (30%)	BALANCE (70%)	TOTAL
Phase 1	R383,400	R894,600	R1,278,000
Phase 2	R207,900	R485,100	R693,000
Phase 3	R240,300	R560,700	R801,000
Phase 4	R218,700	R510,300	R729,000
Phase 5	R264,600	R617,400	R882,000
Phase 6	R229,500	R535,500	R765,000
Total	R1,544,400	R3,603,600	R5,148,000

Billing Model

- **30% deposit** payable on scope approval for each phase — triggers commencement
- **Balance** payable monthly based on tracked hours until phase completion
- **Early completion** results in earlier final payment — MW-GS benefits from faster delivery
- **No phase requires commitment to later phases** — proceed phase-by-phase based on market validation

RECOMMENDED MVP STRATEGY

The full platform is R5.1M. We recommend starting with **Phase 1 + Phase 2** as the MVP package — this delivers the core value proposition and differentiating feature set for **R1,971,000**.

Why Phase 1 + 2 is the Right MVP

WHAT YOU GET	WHY IT MATTERS
Contractor registration & KYC	Qualified, verified users from day one
Company profiles with compliance tracking	B2B credibility engine
Full project management with milestones	The core value loop — contractors managing real projects
AI-powered verification	The differentiator — no other platform offers automated compliance verification
3rd Party Expert verification	Enterprise-grade trust for bankable projects
Tier progression (Bronze → Platinum)	Gamified engagement that drives platform stickiness
Gold Standard certification	The "Gold Standard" brand that banks and insurers will recognise

What You Validate Before Committing to Phase 3+

- Do contractors actually load and manage projects on the platform?
- Does the verification engine deliver accurate compliance assessments?
- Does the tier system drive engagement and retention?
- Is there genuine demand for the marketplace and O&M features?

MVP Investment

ITEM	DETAIL
Phases	Phase 1 + Phase 2
Hours	876
Duration	~11 weeks
Cost	R1,971,000 excl. VAT
Deposit	R591,300 (30%)

ONGOING COSTS – APEX LICENSING & HOSTING

Beyond the development investment, the SEE platform incurs monthly operating costs for APEX module licensing and cloud infrastructure hosting.

APEX Module Licensing (Monthly)

APEX MODULE	USED IN	MONTHLY LICENCE
Forge	Admin dashboard, platform management	Included with deployment
Echo	Notification delivery (email, push, SMS)	R4,500/month
Quantum	AI Verification, Prescriptive Maintenance, SEE.AI	R12,500/month
Zenith	Inverter data ingestion, portfolio knowledge base	R7,500/month
Prism	PDF/report generation, certificate output	R4,500/month
Vector	Verification integrity, AI content grounding	R4,500/month
Total (all modules)		R33,500/month

Licensing fees scale with the platform phases deployed. Phase 1-2 MVP requires only Forge + Echo + Quantum + Prism = R21,500/month. Full licensing applies when all six phases are live.

Cloud Infrastructure Hosting (Estimated Monthly)

COMPONENT	ESTIMATED MONTHLY COST
Compute (application servers, API)	R8,000 - R15,000
Database (RDS/managed PostgreSQL)	R4,000 - R8,000
Storage (S3 — documents, images)	R1,500 - R4,000
CDN & networking	R1,500 - R3,000
AI/LLM API costs (OpenAI/Anthropic)	R5,000 - R20,000
Monitoring & logging	R1,000 - R2,000
Estimated range	R21,000 - R52,000/month

Hosting costs scale with user volume and AI usage. Lower bound assumes early traction (~100 active users); upper bound assumes growth (~1,000+ active users with heavy AI usage).

Total Monthly Operating Cost

SCENARIO	APEX LICENSING	HOSTING (EST.)	TOTAL MONTHLY
MVP (Phase 1-2)	R21,500	R21,000	~R42,500/month
Full Platform	R33,500	R35,000	~R68,500/month
Scale (1,000+ users)	R33,500	R52,000	~R85,500/month



APEX MODULE MAP – WHAT ACCELERATES WHAT

APEX MODULE	SEE PLATFORM CAPABILITY	BUILD REDUCTION
Forge	Admin dashboard, user management, KYC queue, system configuration, reporting	~40% admin panel build saved
Echo	All notifications — milestone updates, O&M alerts, compliance reminders, payment confirmations, marketplace messages	Eliminates custom notification build
Quantum	AI Verification Agent, SEE.AI Assistant, Prescriptive Maintenance, NewsFeed curation, Project Valuation, WEG recommendations	Enterprise AI without building from scratch
Zenith	Multi-brand inverter data ingestion, portfolio knowledge base for SEE.AI	Production-grade data pipeline
Prism	Gold Standard certificates, O&M reports, invoices, Company Profiles, Proposals, Portfolios	Branded PDF generation engine
Vector	AI verification grounding, content generation quality assurance	Hallucination prevention

The Commercial Principle

- **MW-GS/Phoenix Energy does not own APEX** — Reisiger retains all IP
- **SEE accesses APEX capabilities** through a licensing model baked into project delivery
- **SEE benefits from shared R&D** — improvements to APEX modules flow to SEE without additional development cost
- **No infrastructure burden** — Reisiger operates and maintains the APEX platform
- **Faster time-to-market** — enterprise-grade components are available from day one

TECHNOLOGY STACK

LAYER	TECHNOLOGY	RATIONALE
Frontend	React / Next.js	Modern component architecture, SSR for SEO, responsive design
Backend API	Node.js / TypeScript	Type-safe API development, strong ecosystem, shared language with frontend
Database	PostgreSQL (RDS)	Relational integrity for financial transactions, JSONB for flexible schemas
Cache	Redis	Session management, real-time data caching, rate limiting
File Storage	AWS S3	Document storage, compliance artifacts, media assets
Search	Elasticsearch	Full-text search across marketplace listings, projects, and content
Message Queue	AWS SQS / SNS	Async processing — notifications, verification queue, data pipeline
AI/LLM	Anthropic Claude / OpenAI	SEE.AI Assistant, verification engine, content generation
Payments	PayFast	SA payment gateway — card, EFT, instant EFT
Hosting	AWS (af-south-1)	Cape Town region — data sovereignty, low latency for SA users
CI/CD	GitHub Actions	Automated build, test, and deployment pipeline
Monitoring	CloudWatch + Sentry	Infrastructure monitoring, application error tracking

TIMELINE OVERVIEW

,

Week 1 ————— 7 Phase 1: Platform Foundation & Core MVP

- |— Infrastructure, Auth, KYC
- |— Company Profiles, Project Management
- |— Dashboard, Admin Panel, Onboarding

Week 8 ————— 11 Phase 2: Verification Engine & Tier System

- |— AI Verification Agent
- |— 3rd Party Expert Verification
- |— Tier Progression, Gold Standard

Week 12 ————— 16 Phase 3: Service & Hardware Marketplaces

- |— Service Marketplace (5 categories)
- |— Hardware Marketplace + Platform Fees

Week 17 ————— 20 Phase 4: Token Economy, Wallet & Payments

- |— Token Wallet & Purchase Flow
- |— PayFast Integration
- |— Subscription Billing, Rewards Engine

Week 21 ————— 25 Phase 5: O&M Monitoring & Prescriptive AI

- |— Multi-Brand Inverter Integration
- |— Plant Monitoring Dashboards
- |— AI Prescriptive Maintenance

Week 26 ————— 29 Phase 6: SEE.AI & Advanced Intelligence

- |— SEE.AI Assistant (Full Portfolio AI)
- |— AI NewsFeed, Content Generators
- |— Sell My Project, WEG Intelligence

,

Total calendar time: ~29 weeks (~7 months) for full platform delivery

MVP (Phase 1+2): ~11 weeks (~3 months) to market-ready product

RISK CONSIDERATIONS

RISK	IMPACT	MITIGATION
Scope expansion within phases	Cost overrun, timeline slip	Each phase has defined scope. Changes managed through formal change request process.
Inverter API availability	Phase 5 delayed if API access blocked	Engage inverter brands early. Start with WEG (existing relationship). Build adapter pattern for easy brand addition.
Token economy balance	Unsustainable earn/spend ratio	Model token economics before Phase 4 build. Configurable rates allow adjustment without code changes.
Payment gateway compliance	Regulatory delays for marketplace payments	Begin PayFast merchant application during Phase 2. Marketplace payments are regulated — start compliance early.
AI verification accuracy	Compliance decisions based on incorrect AI output	AI verification is advisory (not determinative) in early phases. 3rd Party Expert verification available as Gold Standard. Accuracy improves with training data from real verifications.
Multi-brand data inconsistency	Inverter APIs return different data at different intervals	Data normalisation layer (Phase 5) abstracts brand differences. Missing data points handled gracefully with "data unavailable" indicators.
Market adoption risk	Platform built but contractors don't adopt	Phase 1+2 MVP strategy validates core loop before committing to full build. ~R2M validates the market before ~R5M is committed.

WHAT MW-GS / PHOENIX ENERGY PROVIDES

For each phase, the following inputs are required:

Phase 1 (MVP)

- Hosting account credentials (AWS or Reisiger-managed — to be agreed)
- Phoenix Energy Solutions design review participation (2-3 sessions)
- KYC requirements specification — what constitutes valid CIPC, VAT, and Director ID verification
- Milestone template definitions — what milestones apply to Development, Construction, and O&M phases

Phase 2 (Verification)

- Compliance rule definitions — what makes a milestone submission compliant, per milestone type
- Expert verification criteria — what qualifies a 3rd party expert, what the verification form looks like
- Tier progression rules — how many compliant projects to advance from Bronze to Silver, Silver to Gold, etc.

Phase 3 (Marketplaces)

- Initial service provider partners for marketplace launch
- Hardware supplier partnerships or affiliate agreements
- Fee structure decisions — listing fees, platform fee percentages per tier

Phase 4 (Token Economy)

- Token pricing decision — fiat-to-token exchange rate
- Cashback rates per tier
- PayFast merchant account (Reisiger can assist with application)

Phase 5 (O&M)

- API credentials for at least one inverter brand (WEG preferred as starting point)
- O&M licence pricing structure — Basic/Premium/AI rates per project size category
- Sample installation data for testing

Phase 6 (AI)

- WEG product catalogue and technical specifications
- Approval of AI assistant personality, tone, and behavioural guidelines
- Languages required for multi-language support

NEXT STEPS

1. **Align on MVP scope** — Confirm Phase 1 + Phase 2 as the initial delivery, or adjust phasing based on MW-GS/Phoenix Energy priorities
2. **Schedule architecture session** — Reisiger, MW-GS (Marcello), and Phoenix Energy Solutions (Erin, Ricardo) to review this proposal and finalise MVP scope boundaries
3. **Finalise commercial terms** — Confirm rate, deposit structure, and billing cycle
4. **Phase 1 kickoff** — Upon scope approval and deposit receipt, mobilise engineering team
5. **Ongoing governance** — Weekly progress reports, fortnightly stakeholder demos, monthly steering committee

ENGAGEMENT TEAM

ROLE	PERSON	ORGANISATION
Account Owner	Basson Engelbrecht	Reisiger (CEO)
Technical Lead	Christiaan Oosthuizen	Reisiger (CTO)
Engineering	2 dedicated engineers	Reisiger
Client Sponsor	Marcello Pompa	MW-GS (CEO)
Domain Partners	Erin & Ricardo	Phoenix Energy Solutions

Prepared by Reisiger Engineering / April 2026 / v1.0

This proposal is valid for 60 days from the date above.